DO NOW — RECALL

Binary search: trace and code

Name: _____ Class: _____ Date: _____

1. Does linear search require ordered input?

2. What must be true of the two input runs when merging them?

3. For seven elements, what is the last zero-based index?

4. Find the integer midpoint of low=2 and high=8.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

No. It scans sequentially and works on unsorted arrays.

2. Model answer

They must already be sorted.

3. Model answer

6.

4. Model answer

5: 2 + (8-2)/2 = 5.

Vocabulary — know these terms

precondition — Input assumption: the array is ordered.

low — Lower inclusive search index.

high — Upper inclusive search index.

mid — Index of the middle candidate.

O(log n) — Search repeatedly halves the candidate set.

NEW LEARNING

Read, annotate and explain

Sort before searching

Binary search relies on a sorted array; otherwise discarding half may lose the target.

Inspect the middle

Compare target to the midpoint and keep only the half that could contain it.

Stop correctly

Return an index when found; return -1 if low becomes greater than high.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Search for 19 in [3,7,10,15,19,23,27]. Record every midpoint checked.

Predict first

Use inclusive low and high indexes.

Model solution

mid 3 → 15, so low=4; mid 5 → 23, so high=4; mid 4 → 19. Return 4.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Search for 8 in [2,5,8,11,14]. List midpoint indexes.

Question 2

Search for 13 in [2,4,6,8,10,12,14]. List indexes checked and final return.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why setting low = mid instead of mid + 1 can cause failure.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Binary search is always safe on an unsorted array.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.